# Advanced Applied Statistics

## Lecture 6: Survival Analysis and Expectation Maximisation

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30 September 2025

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# Learning Objectives of This Lecture

- Explain the principles of survival analysis
- Explain the concept of hazard rate and hazard ratio
- Indenify cases where Kaplan–Meier estimate, the log-rank test, and Cox's proportional hazard model are applicable.
- Apply survival analysis methods in real problems
- Explain the steps of Expectation Maximisation algorithm
- Apply and implement EM algortihm in applications

**Table 9.1** *Insurance company life table; at each age, $n$ = number of policy holders, $y$ = number of deaths, $\hat{h}$ = hazard rate $y/n$, $\hat{S}$ = survival probability estimate* (9.6).

| Age | $n$ | $y$ | $\hat{h}$ | $\hat{S}$ | Age | $n$ | $y$ | $\hat{h}$ | $\hat{S}$ |
|---|---|---|---|---|---|---|---|---|---|
| 30 | 116 | 0 | .000 | 1.000 | 60 | 231 | 1 | .004 | .889 |
| 31 | 44 | 0 | .000 | 1.000 | 61 | 245 | 5 | .020 | .871 |
| 32 | 95 | 0 | .000 | 1.000 | 62 | 196 | 5 | .026 | .849 |
| 33 | 97 | 0 | .000 | 1.000 | 63 | 180 | 4 | .022 | .830 |
| 34 | 120 | 0 | .000 | 1.000 | 64 | 170 | 2 | .012 | .820 |
| 35 | 71 | 1 | .014 | .986 | 65 | 114 | 0 | .000 | .820 |
| 36 | 125 | 0 | .000 | .986 | 66 | 185 | 5 | .027 | .798 |
| 37 | 122 | 0 | .000 | .986 | 67 | 127 | 2 | .016 | .785 |
| 38 | 82 | 0 | .000 | .986 | 68 | 127 | 5 | .039 | .755 |
| 39 | 113 | 0 | .000 | .986 | 69 | 158 | 2 | .013 | .745 |
| 40 | 79 | 0 | .000 | .986 | 70 | 100 | 3 | .030 | .723 |
| 41 | 90 | 0 | .000 | .986 | 71 | 155 | 4 | .026 | .704 |

# Discrete Case

- Let $X$ a typical life time,
$$f_i = P(X = i) \tag{1}$$
is the probability of **dying at age** $i$.

- Further,
$$S_i = \sum_{j \geq i} f_j = P(X \geq i) \tag{2}$$
is the probability of **surviving past age** $i - 1$.

- The **hazard rate at age** $i$ is the probability of dying at age $i$ given one survived past age $i - 1$, or,
$$h_i = f_i / S_i = P(X = i | X \geq i). \tag{3}$$

- Probability of surviving past age $j$ given surviving past age $i - 1$
$$S_{ij} = \prod_{k=i}^{j} (1 - h_k) = P(X > j | X \geq i). \tag{4}$$

# Continuous Case

- We replace the discrete variable $X$ with continuous time $T$ with the density function $f(t)$.
- The reverse cdf or **survival function**

$$S(t) = P(T \geq t) = \int_t^{\infty} f(x)dx \tag{5}$$

- The **hazard rate**

$$h(t) = \frac{f(t)}{S(t)} \tag{6}$$

- Now

$$h(t)dt = P(T \in (t, t+dt)|T \geq t) \tag{7}$$

  for $dt \to 0$.

- Moreover,

$$P(T \geq t_1|T \geq t_0) = \exp\left(-\int_{t_0}^{t_1} h(x)dx\right). \tag{8}$$

# Exponential Density for Lifetime

## Example

Assume one-sided exponential density

$$f(t) = (1/c)e^{-t/c} \quad t \geq 0 \tag{9}$$

The survial function $S(t) = e^{-t/c}$. The hazard rate thus is $h(t) = 1/c$ (constant).

## Exercise

The exponential density is often referred to as 'memoryless' – why?

# Censored data

- Clinical trial where two treatments are compared: (A) Chemotherapy and (B) Chemotherapy plus radiotherapy
- A + sign indicates that the patient was lost to followup (i.e., they lived longer).

**Table 9.2** *Censored survival times in days, from two arms of the* NCOG *study of head/neck cancer.*

Arm A: Chemotherapy

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 7 | 34 | 42 | 63 | 64 | 74+ | 83 | 84 | 91 |
| 108 | 112 | 129 | 133 | 133 | 139 | 140 | 140 | 146 |
| 149 | 154 | 157 | 160 | 160 | 165 | 173 | 176 | 185+ |
| 218 | 225 | 241 | 248 | 273 | 277 | 279+ | 297 | 319+ |
| 405 | 417 | 420 | 440 | 523 | 523+ | 583 | 594 | 1101 |
| 1116+ | 1146 | 1226+ | 1349+ | 1412+ | 1417 | | | |

Arm B: Chemotherapy+Radiation

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 37 | 84 | 92 | 94 | 110 | 112 | 119 | 127 | 130 |
| 133 | 140 | 146 | 155 | 159 | 169+ | 173 | 179 | 194 |
| 195 | 209 | 249 | 281 | 319 | 339 | 432 | 469 | 519 |
| 528+ | 547+ | 613+ | 633 | 725 | 759+ | 817 | 1092+ | 1245+ |
| 1331+ | 1557 | 1642+ | 1771+ | 1776 | 1897+ | 2023+ | 2146+ | 2297+ |

# The Data Binned by Month

**Table 9.3** `Arm_A` *of the* `NCOG` *head/neck cancer study, binned by month; $n$ = number at risk, $y$ = number of deaths, $l$ = lost to followup, $h$ = hazard rate $y/n$; $\hat{S}$ = life table survival estimate.*

| Month | $n$ | $y$ | $l$ | $h$ | $\hat{S}$ | Month | $n$ | $y$ | $l$ | $h$ | $\hat{S}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 51 | 1 | 0 | .020 | .980 | 25 | 7 | 0 | 0 | .000 | .184 |
| 2 | 50 | 2 | 0 | .040 | .941 | 26 | 7 | 0 | 0 | .000 | .184 |
| 3 | 48 | 5 | 1 | .104 | .843 | 27 | 7 | 0 | 0 | .000 | .184 |
| 4 | 42 | 2 | 0 | .048 | .803 | 28 | 7 | 0 | 0 | .000 | .184 |
| 5 | 40 | 8 | 0 | .200 | .642 | 29 | 7 | 0 | 0 | .000 | .184 |
| 6 | 32 | 7 | 0 | .219 | .502 | 30 | 7 | 0 | 0 | .000 | .184 |
| 7 | 25 | 0 | 1 | .000 | .502 | 31 | 7 | 0 | 0 | .000 | .184 |
| 8 | 24 | 3 | 0 | .125 | .439 | 32 | 7 | 0 | 0 | .000 | .184 |
| 9 | 21 | 2 | 0 | .095 | .397 | 33 | 7 | 0 | 0 | .000 | .184 |
| 10 | 19 | 2 | 1 | .105 | .355 | 34 | 7 | 0 | 0 | .000 | .184 |
| 11 | 16 | 0 | 1 | .000 | .355 | 35 | 7 | 0 | 0 | .000 | .184 |
| 12 | 15 | 0 | 0 | .000 | .355 | 36 | 7 | 0 | 0 | .000 | .184 |
| 13 | 15 | 0 | 0 | .000 | .355 | 37 | 7 | 1 | 1 | .143 | .158 |
| 14 | 15 | 3 | 0 | .200 | .284 | 38 | 5 | 1 | 0 | .200 | .126 |
| 15 | 12 | 1 | 0 | .083 | .261 | 39 | 4 | 0 | 0 | .000 | .126 |
| 16 | 11 | 0 | 0 | .000 | .261 | 40 | 4 | 0 | 0 | .000 | .126 |

# The Kaplan–Meier Non-Parametric Estimate

- The observations in censored data problems are of the form $z_i = (t_i, d_i)$, where $t_i$ is the observed survial time and $d_i$ is binary with unity if death is observed, zero otherwise.
- Let $t_{(1)} < t_{(2)} < ... < t_{(n)}$ be the ordered survival times. The **Kaplan–Meier** formula estimates the **survival probability** as

$$\hat{S}_{(j)} = \prod_{k \leq j} \left( \frac{n-k}{n-k+1} \right)^{d_{(k)}} \tag{10}$$

- The accuracy estimate (**Greenwood's formula**)

$$\sigma_{\mathrm{KM}} = \hat{S}_{(j)} \sqrt{\sum_{k \leq j} \frac{y_k}{n_k (n_k - y_k)}} \tag{11}$$

# Kaplan–Meier Survival Curves

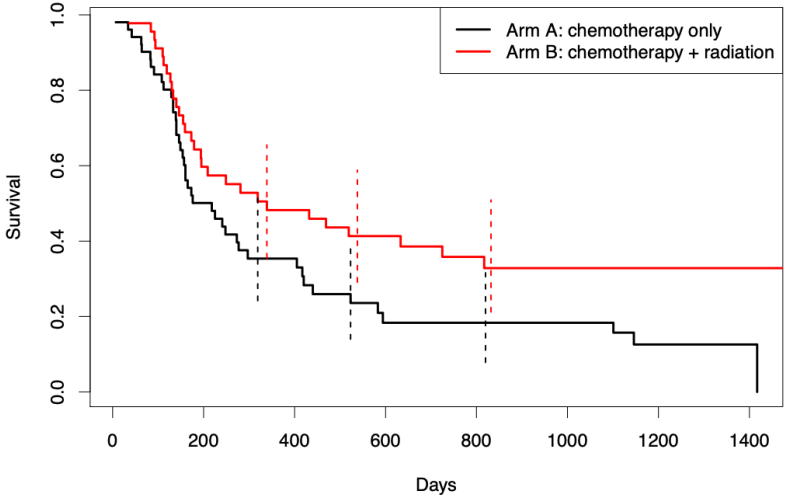



**Figure 9.1** `NCOG` Kaplan–Meier survival curves; lower `Arm_A` (chemotherapy only); upper `Arm_B` (chemotherapy+radiation). Vertical lines indicate approximate 95% confidence intervals.

# Hazard Ratio

## Definition

The **hazard ratio** (HR) is the ratio of the hazard rates corresponding to the conditions characterised by two distinct levels of a treatment variable of interest.

## Example

In a clinical study of a drug, the treated population dies at twice the rate per unit time when compared to the control population, that is

$$\mathrm{HR} = \frac{h_{\text{treated}}}{h_{\text{control}}} = 2, \tag{12}$$

which indicates a higher hazard of death from the treatment.

# Hazard Ratio - Parametric example

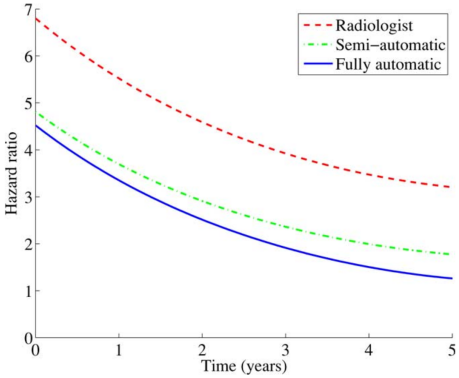
From Pedersen et al. 2012.

## Example

- CVD patients were divided into two equally sized groups by the AC24 score (therhold at 3.5).
- For each group, the pdf and the cdf of the survival time estimated by a Gaussian kernel density estimator.
- From the kernel estimates, the hazard functions and the hazard ratio were estimated.
- Conclusions: (1) the manual scores are better than the automated one (2) there is an increased CVD death risk in the group of higher AC24 scores.

TABLE VI
HAZARD RATIOS FOR THE EPIPF BASELINE STUDY

| | $R(0)$ | $\frac{1}{5\mathrm{y}}\int_0^{5\mathrm{y}} R(t)dt$ |
|---|---|---|
| Radiologist | 6.8 | 4.5 |
| This study ($\mathrm{Semi_{bl}}$) | 4.8 | 2.8 |
| This study ($\mathrm{Auto_{bl}}$) | 4.5 | 2.4 |



# Log-Rank Test

- Target: a non-parametric two-sample test for **censored** survival data (clinical trials)

**Table 9.4** *Life table comparison for the first six months of the* `NCOG` *study. For example, at the beginning of the sixth month after treatment, there were 33 remaining* `Arm_B` *patients, of whom 4 died during the month, compared with 32 at risk and 7 dying in* `Arm_A`*. The conditional expected number of deaths in* `Arm_A`*, assuming the null hypothesis of equal hazard rates in both arms, was 5.42, using expression* (9.24).

| Month | `Arm_B` | | `Arm_A` | | Expected number `Arm_A` deaths |
|---|---|---|---|---|---|
| | At risk | Died | At risk | Died | |
| 1 | 45 | 0 | 51 | 1 | .53 |
| 2 | 45 | 1 | 50 | 2 | 1.56 |
| 3 | 44 | 1 | 48 | 5 | 3.13 |
| 4 | 43 | 5 | 42 | 2 | 3.46 |
| 5 | 38 | 5 | 40 | 8 | 6.67 |
| 6 | 33 | 4 | 32 | 7 | 5.42 |

# Log-Rank Test: Forming the Test Statistic

**Table 9.5** *Two-by-two display of month-6 data for the* **NCOG** *study. $E$ is the expected number of* **Arm_A** *deaths assuming the null hypothesis of equal hazard rates (last column of Table 9.4).*

| | Died | Survived | |
|---|---|---|---|
| **Arm_A** | $y = 7$<br>$E = 5.42$ | 25 | $n_A = 32$ |
| **Arm_B** | 4 | 29 | $n_B = 33$ |
| | $n_d = 11$ | $n_s = 54$ | $n = 65$ |

Under the null hypothesis at month 6,

- the hazard rates are equal in groups $A$ and $B$
- the deaths $y$ are hypergeometrically distributed with the mean $\mu = \frac{n_A n_d}{n}$ and variance $\sigma^2 = \frac{n_A n_B n_d n_s}{n^2(n-1)}$

# Log-Rank Test: Forming the Test Statistic (2)

- Similar two-by-two table can be formed for all the $N$ months.
- The **log-rank test statistic** $Z$ is defined as

$$Z = \frac{\sum_{i=1}^{N}(y_i - \mu_i)}{\sqrt{\sum_{i=1}^{N}\sigma_i^2}} \tag{13}$$

- Asymptotically, on the basis of the central limit theorem,

$$Z \sim N(0, 1) \tag{14}$$

under the null hypothesis of equal hazard rates on each month.

# Proportional Hazards Model

- Goal: **full regression analysis of censored data**
- Individual points

$$z_i = (c_i, t_i, d_i) \tag{15}$$

where $c_i$ is (row) vector of covariates whose effect on survival we want to assess.

**Table 9.6** *Pediatric cancer data, first 20 of 1620 children.* **Sex** *1 = male, 2 = female;* **race** *1 = white, 2 = nonwhite;* **age** *in years;* **entry** *= calendar date of entry in days since July 1, 2001;* **far** *= home distance from treatment center in miles;* **t** *= survival time in days;* **d** *= 1 if death observed, 0 if not.*

| sex | race | age | entry | far | t | d |
|---|---|---|---|---|---|---|
| 1 | 1 | 2.50 | 710 | 108 | 325 | 0 |
| 2 | 1 | 10.00 | 1866 | 38 | 1451 | 0 |
| 2 | 2 | 18.17 | 2531 | 100 | 221 | 0 |
| 2 | 1 | 3.92 | 2210 | 100 | 2158 | 0 |
| 1 | 1 | 11.83 | 875 | 78 | 760 | 0 |
| 2 | 1 | 11.17 | 1419 | 0 | 168 | 0 |
| 2 | 1 | 5.17 | 1264 | 28 | 2976 | 0 |

# Proportional Hazards Model (2)

- For the Hazard rate, for individual $i$, it is assumed

$$h_i(t) = h_0(t)\theta_i \tag{16}$$

  where $\theta_i = e^{c_i'\beta}$, and $h_0(t)$ is a **baseline** hazard rate.

- It follows

$$S_i(t) = S_0(t)^{\theta_i}. \tag{17}$$

- Let J be the number of observed deaths. Assume an **ordering** of the death occurences so that

$$T_{(1)} < T_{(2)} < \ldots < T_{(J)}. \tag{18}$$

- Just before time $T_{(j)}$ there is a **risk set** of individuals under observation

$$R_j = \{i : t_i \geq T_{(j)}\}. \tag{19}$$

# Proportional Hazards Model (3)

## Lemma

*Under the proportional hazards model, the conditional probability, given the risk set $R_j$ that individual $i$ in $R_j$ is the one observed to die at time $T_{(j)}$ is*

$$P(i_j = i | R_j) = \frac{e^{c_i' \beta}}{\sum_{k \in R_j} e^{c_k' \beta}}. \tag{20}$$

*In other words, given that one person dies at time $T_{(j)}$, the probability of individuyal $i$ is proportional to $e^{c_i' \beta}$ among the set of individuals at risk.*

# Partial Likelihood

- To esetimate the parameter vector $\beta$ one constructs the partial likelihood

$$L(\beta) = \prod_{j=1}^{J} \frac{e^{c_i'\beta}}{\sum_{k \in R_j} e^{c_k'\beta}} \tag{21}$$

- The MLE and parameter covariance matric estimate are obtained as the minimiser of the negative log-likelihood and the inverse Fisher information matrix, respectively.

# Proportional Hazards Analysis with Pediatric cancer data

## Example

**Table 9.7** *Proportional hazards analysis of pediatric cancer data (`age`, `entry` and `far` standardized). `Age` significantly negative, older children doing better; `entry` very significantly negative, showing hazard rate declining with calendar date of entry; `far` very significantly positive, indicating worse results for children living farther away from the treatment center. Last two columns show limits of approximate 95% confidence intervals for* $\exp(\beta)$.

| | $\beta$ | sd | $z$-value | $p$-value | $\exp(\beta)$ | Lower | Upper |
|---|---|---|---|---|---|---|---|
| `sex` | −.023 | .160 | −.142 | .887 | .98 | .71 | 1.34 |
| `race` | .282 | .169 | 1.669 | .095 | 1.33 | .95 | 1.85 |
| `age` | −.235 | .088 | −2.664 | .008 | .79 | .67 | .94 |
| `entry` | −.460 | .079 | −5.855 | .000 | .63 | .54 | .74 |
| `far` | .296 | .072 | 4.117 | .000 | 1.34 | 1.17 | 1.55 |

# Quiz

# Sampling from a Gaussian Mixture

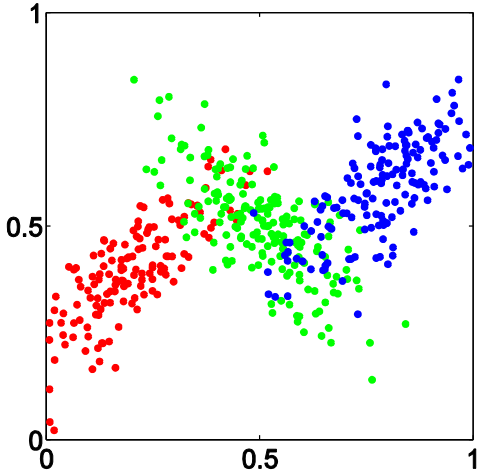
**In simulation, the mixture parameters are known**

- To generate a data point:

  First draw one of the components *k* with probabilies $p(k)$

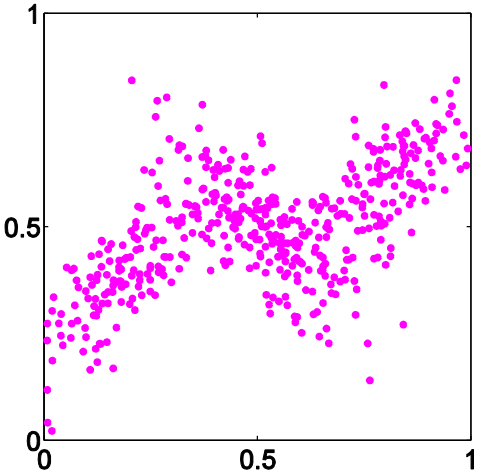
  then draw a sample $\mathbf{x}^{(n)}$ from the component $p(\mathbf{x}|k)$
- Repeat these two steps for each new data point

# Synthetic Data Set



# Synthetic Data Set Without Labels

**Can you say from which cluster each data point comes from?**



# Fitting the Gaussian Mixture

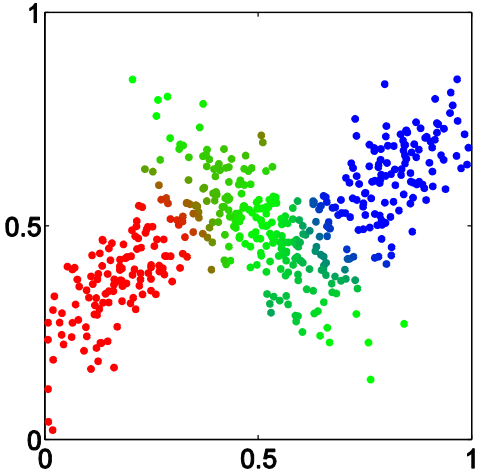
- *Inverse* of the simulation process: given the data set, find the mixture parameters
  - mixing coefficients
  - means
  - covariances
- *If we knew* which component generated each data point, the maximum likelihood solution would be trivial
  - Separate Fitting Each Component to the corresponding cluster
- However, the data set is generally unlabelled
- The labels are referred to as the **latent** (= hidden) **variables**

# Posterior Probabilities

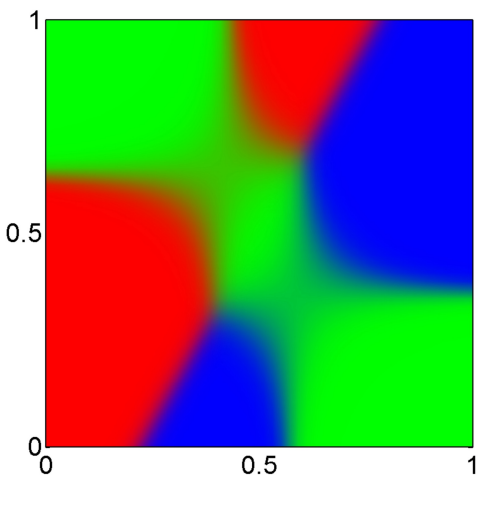
- We can think of the mixing coefficients as prior probabilities for the components
- For a given value of $\mathbf{x}$ we can evaluate the corresponding posterior probabilities, called *responsibilities*
- These are given by Bayes' theorem as

$$\gamma_k(\mathbf{x}) \equiv p(k|\mathbf{x}) = \frac{p(k)p(\mathbf{x}|k)}{p(\mathbf{x})} = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

# Posterior Probabilities (colour coded)



# Posterior Probability Map



# Maximum Likelihood Estimation for the GMM

- The log likelihood function takes the form

$$\ln p(D|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

- Note: sum over components appears *inside* the log
- There is *no closed form solution* for maximum likelihood

# Questions and Answers

Q: How to maximize the log likelihood

A: Expectation-Maximization (EM) algorithm

Q: How to avoid singularities in the likelihood function

A1: Good initialization helps

A2: Bayesian treatment (beyond the scope of this course)

Q: How to choose number of components *K*

A: Bayesian treatment (beyond the scope of this course)

# EM Algorithm – Informal Derivation

- By differentiating the log likelihood, and setting the derivative with respect to $\boldsymbol{\mu}_j$ equal to zero yields

$$-\sum_{n=1}^{N} \underbrace{\frac{\pi_j \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}{\sum_k \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}}_{\gamma_j(\mathbf{x}_n)} \boldsymbol{\Sigma}_j^{-1}(\mathbf{x}_n - \boldsymbol{\mu}_j) = 0$$

and further

$$\boldsymbol{\mu}_j = \frac{\sum_{n=1}^{N} \gamma_j(\mathbf{x}_n)\mathbf{x}_n}{\sum_{n=1}^{N} \gamma_j(\mathbf{x}_n)}$$

which is the **weighted mean of the data**

# EM Algorithm – Informal Derivation

- Similarly for the covariances

$$\boldsymbol{\Sigma}_j = \frac{\sum_{n=1}^{N} \gamma_j(\mathbf{x}_n)(\mathbf{x}_n - \boldsymbol{\mu}_j)(\mathbf{x}_n - \boldsymbol{\mu}_j)^{\mathsf{T}}}{\sum_{n=1}^{N} \gamma_j(\mathbf{x}_n)}$$

- For mixing coefficients use Lagrange multipliers yields

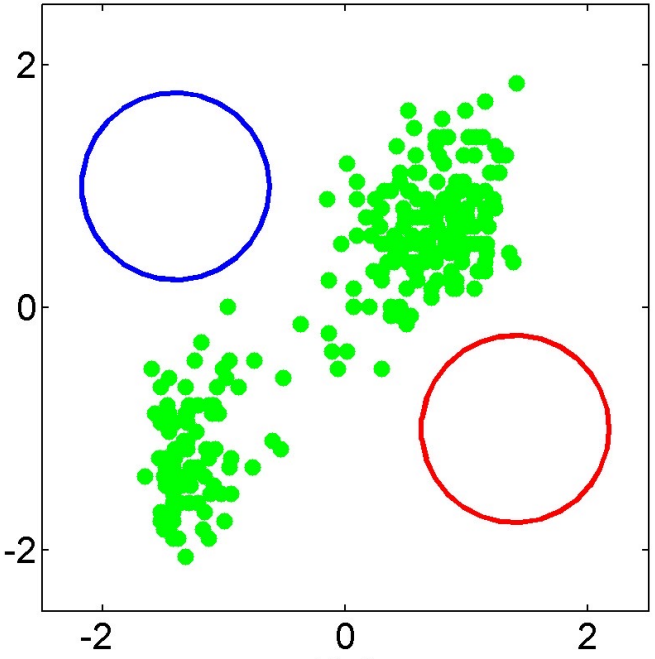
$$\pi_j = \frac{1}{N} \sum_{n=1}^{N} \gamma_j(\mathbf{x}_n)$$

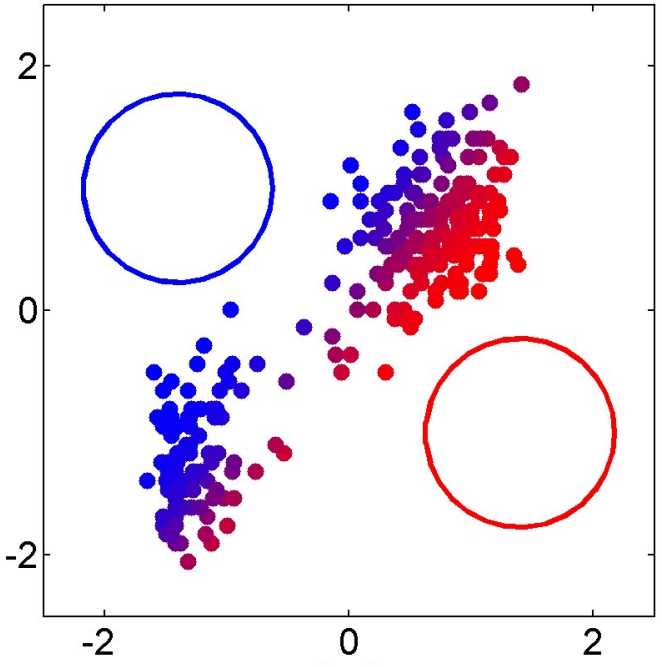
# EM Algorithm – Informal Derivation

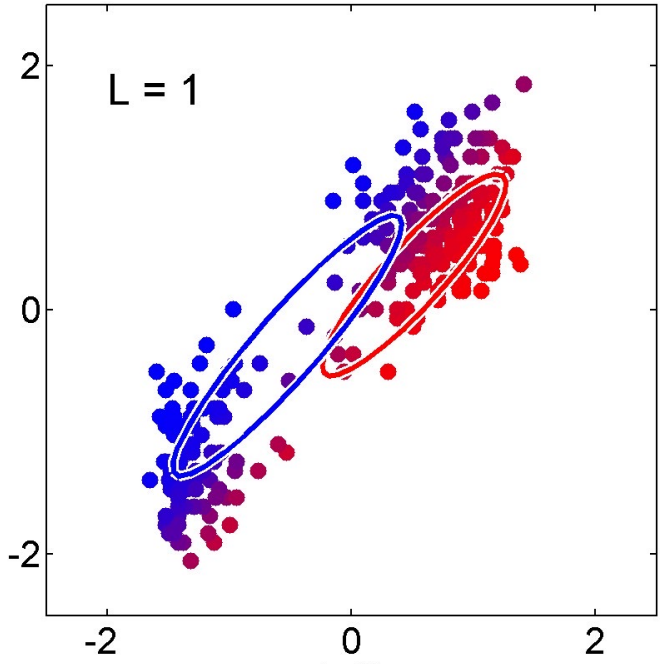
- The solutions are **not** closed form: they are **coupled**
- Suggests an iterative scheme:
  - Make initial guesses for the parameters
  - Alternate between

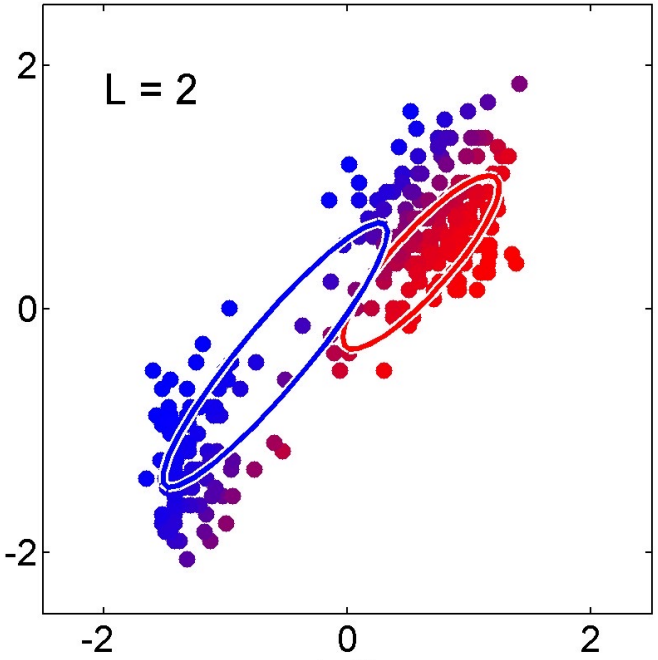
    E-step: evaluate responsibilities

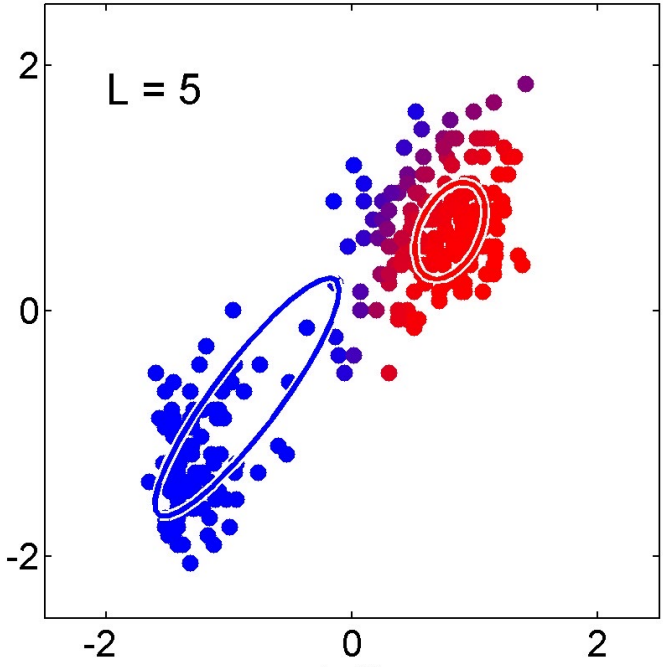
    M-step: update parameters using the ML results

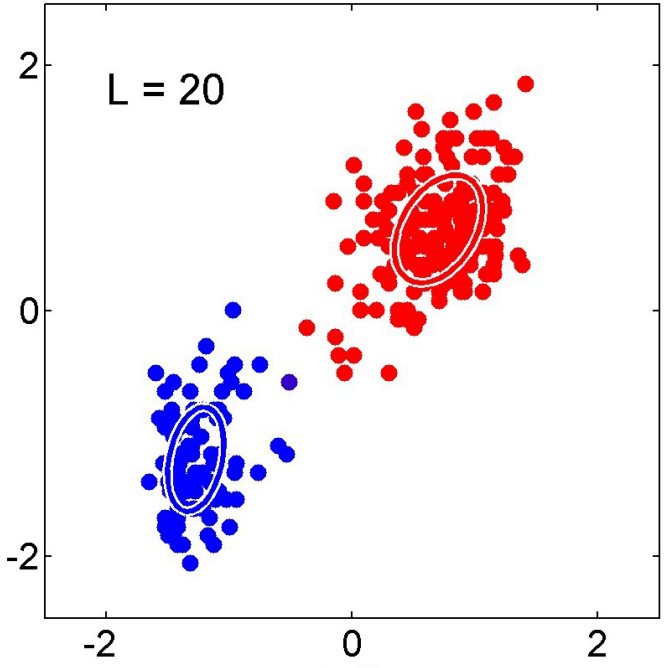

2
0
-2
-2
0
2


2
0
-2
-2
0
2


L = 1
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0
-2
-2
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2


L = 2
2
0
-2
-2
0
2


L = 5
2
0
-2
-2
0
2


L = 20
2
0
-2
-2
0
2

# General Expectation Maximisation

- Let the likelihood of the **complete data** be

$$p\left(\mathbf{X},\mathbf{Z}\middle|\theta\right)$$

where $\mathbf{X}$ is the **observed data** and $\mathbf{Z}$ is **missing data**

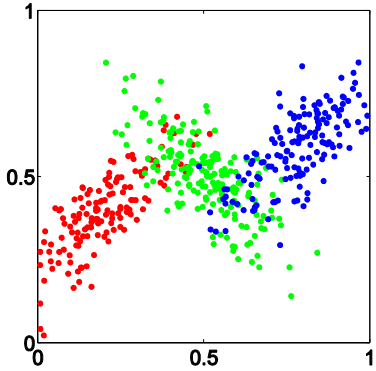
**E-step**: Construct

$$\chi\left(\theta,\hat{\theta}^{(i-1)}\right)=\mathrm{E}_{\mathbf{Z}\middle|\mathbf{X},\theta^{(i-1)}}\left\{\log p\left(\mathbf{X},\mathbf{Z}\middle|\theta\right)\middle|\mathbf{X},\hat{\theta}^{(i-1)}\right\}$$

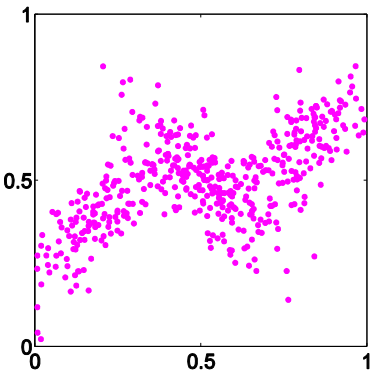
**M-step**: Maximize

$$\hat{\theta}^{(i)}=\arg\max_{\theta}\chi\left(\theta,\hat{\theta}^{(i-1)}\right)$$

# Complete and Incomplete Data



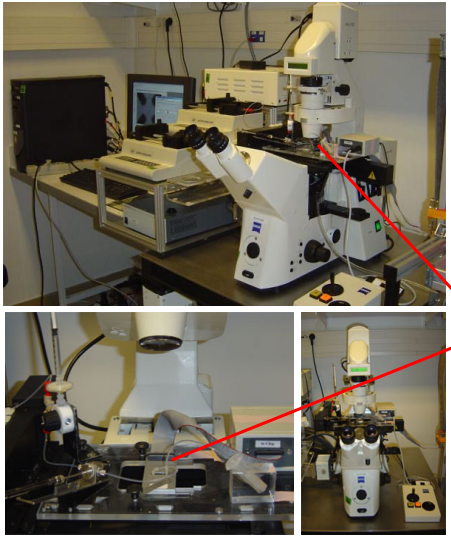

complete

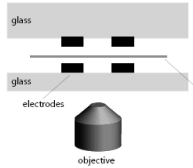


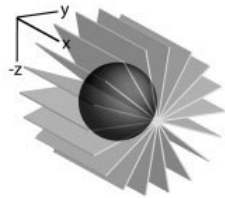

incomplete

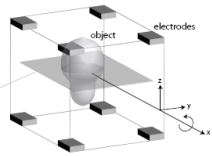
# Application Example of EM

**Micro-rotation fluorescence imaging for live cells**





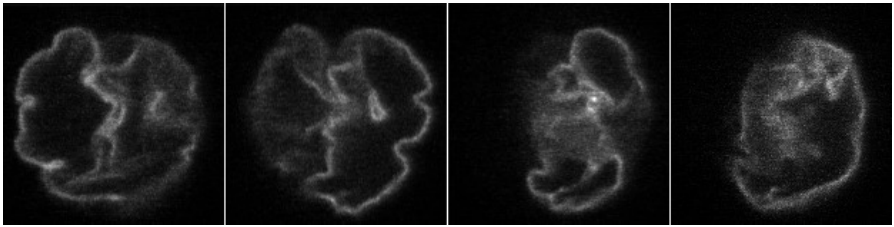






http://www.pfid.org/AUTOMATION/

# Examples of micro-rotation images

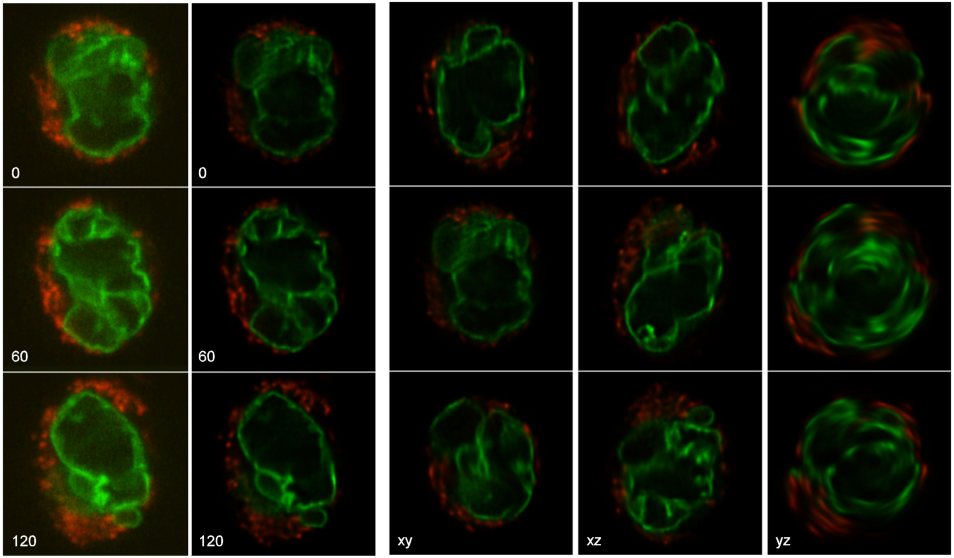


A human living cell, expressing fluorescence at nuclear envelope



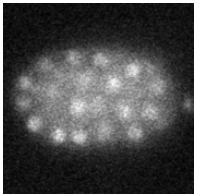
Laksameethanasan et al. 2009

# EMTV lamin-GFP + mito-OFP



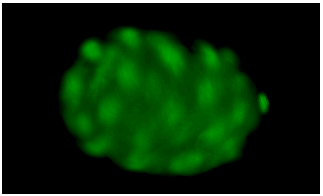
# Example of reconstructions

C.elegans embryo

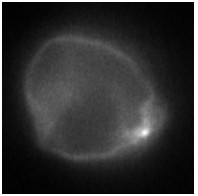


reconstruction

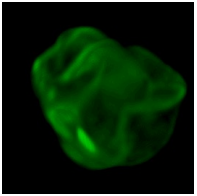




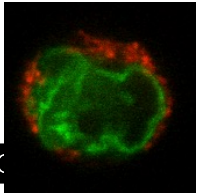
human living cell



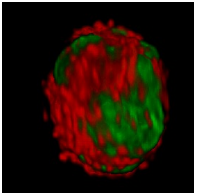




human living cell







# Summary

- K-means is a special case of fitting a Gaussian Mixture Model into data
- EM is the standard way of fitting a Gaussian Mixture Model
- Expectation Maximization is a general principle for solving maximum likelihood estimates with missing data